

Drinking Water - Specification

Fifth Revision

IS 10500 : 2012

1 Scope

1.1 This standard lays down the requirements for drinking water supplied to the consumer. It specifies physical, chemical, organoleptic, bacteriological and radiological requirements that drinking water shall conform to in order to be safe and suitable for human consumption.

1.2 The limits specified in this standard are applicable to all piped and packaged drinking water intended for human consumption, including water treated at household level.

NOTE For water from natural springs and mineral water, refer to IS 13428.

2 References

The following Indian Standards contain provisions which, through reference in this text, constitute provisions of this standard. All standards are subject to revision and parties to agreements based on this standard are encouraged to apply the most recent editions of the standards listed below.

IS 3025 (Part 4) : 1983 - Methods of sampling and test for water and wastewater: Measurement of pH.

IS 3025 (Part 17) : 1984 - Acidity of water.

IS 3025 (Part 21) : 1983 - Hardness.

3 Terminology

3.1 Acceptable Limit - The value below which no adverse health effect is expected for lifetime consumption. Values above the acceptable limit but below the permissible limit may be used if no better source is available.

3.2 Permissible Limit - The value below which water may be used in absence of any better alternative source. Above this limit, the water is considered hazardous to health.

3.3 Essential Characteristic - Any requirement whose non-compliance would render the water unacceptable for human consumption regardless of circumstances.

4 Requirements

4.1 General

4.1.1 Drinking water shall be free from suspended solids, pathogens, toxic chemicals, and radioactive substances at levels that could constitute a hazard to health. The water shall be clear, colourless,

odourless and pleasant in taste.

4.1.2 Emergency Provisions - In emergency situations, some parameters may be relaxed by the competent authority provided adequate public health measures are in place.

4.2 Organoleptic and Physical Characteristics

4.2.1 The organoleptic and physical parameters of drinking water shall conform to the requirements given in Table 1.

4.2.2 Colour shall be determined by the method specified in IS 3025 (Part 4). Turbidity shall be measured in Nephelometric Turbidity Units (NTU).

Table 1 Organoleptic and Physical Requirements (IS 10500:2012)

Sl. No.	Characteristic	Acceptable Limit	Permissible Limit	Method of Test
1	Colour, Hazen units, Max	5	15	IS 3025 (Pt 4)
2	Odour	Agreeable	Agreeable	IS 3025 (Pt 5)
3	Taste	Agreeable	Agreeable	Organoleptic
4	Turbidity, NTU, Max	1	5	IS 3025 (Pt 10)
5	pH value	6.5 to 8.5	No relax.	IS 3025 (Pt 11)
6	Total Dissolved Solids, mg/L Max	500	2000	IS 3025 (Pt 16)
7	Total Hardness as CaCO ₃ , mg/L Max	200	600	IS 3025 (Pt 21)
8	Calcium as Ca, mg/L Max	75	200	IS 3025 (Pt 40)
9	Magnesium as Mg, mg/L Max	30	100	IS 3025 (Pt 46)

4.3 Chemical Requirements - Non-toxic Substances

4.3.1 The chemical (non-toxic) requirements shall conform to the limits specified. Substances listed are not toxic in themselves but can affect palatability or corrode distribution systems if present above the specified limits.

4.3.2 Iron (IS 3025 Part 53) - Max 0.3 mg/L acceptable; 1.0 mg/L permissible. Excessive iron causes discolouration and an unpleasant metallic taste.

4.3.3 Chloride as Cl (IS 3025 Part 32) - Max 250 mg/L acceptable; 1000 mg/L permissible.

5 Bacteriological Requirements

5.1 Treated water entering the distribution system

5.1.1 Escherichia coli or thermotolerant coliform bacteria must not be detectable in any 100 mL sample.

5.1.2 Total coliform bacteria must not be detectable in any 100 mL sample.

5.2 Treated water in the distribution system

5.2.1 E. coli or thermotolerant coliform bacteria must not be detectable in any 100 mL sample.

5.2.2 Coliform bacteria shall not be detectable in 95 percent of samples taken throughout the year in any 12-month period for large supplies.

6 Radiological Requirements

6.1 Gross Alpha Activity - The indicative dose for gross alpha activity shall not exceed 0.1 Bq/litre.

6.2 Gross Beta Activity - The indicative dose for gross beta activity shall not exceed 1.0 Bq/litre.

6.3 Where either of the above screening values is exceeded, a more detailed radiological analysis shall be conducted for specific radionuclides before the water is deemed unsafe.

Table 2 Toxic Substances - Maximum Permissible Concentrations (IS 10500:2012)

Sl. No.	Substance	Max. Permissible Limit (mg/L)	Method of Test (IS 3025)
1	Arsenic (As)	0.01	Part 37
2	Cadmium (Cd)	0.003	Part 41
3	Chromium (Cr-VI)	0.05	Part 52
4	Cyanide (CN)	0.05	Part 27
5	Fluoride (F)	1.0	Part 60
6	Lead (Pb)	0.01	Part 47
7	Mercury (Hg)	0.001	Part 48
8	Nitrate (NO ₃)	45	Part 34
9	Nitrite (NO ₂)	0.02	Part 34
10	Selenium (Se)	0.01	Part 62
11	Barium (Ba)	0.7	Part 63
12	Antimony (Sb)	0.005	Part 64
13	Bromate (BrO ₃)	0.01	Part 65